Credit Risk Prediction Using Machine Learning

Credit Risk Dataset
32,581 customer records
Binary classification: loan_status

- Financial institutions need to identify customers who may not default on loans.
- The goal is to build classification models that predict loan status.
- Compare model performance before and after PCA.
- Improve the best model using GridSearchCV hyperparameter tuning

Problem Statement

The financial services company wants to modernise its credit approval system using historical customer data.

The goal is to classify applicants as higher-risk or lower-risk using supervised machine learning.

The target variable is loan_status, so this is a binary classification problem.

False approvals can lead to financial losses, while false rejections can lead to lost business opportunities.

Project objective

Build and evaluate models that can predict loan status from customer and loan characteristics.

Model comparison

Train Logistic Regression, KNN and Decision Tree before and after PCA.

Optimisation

Tune the selected Decision Tree using GridSearchCV and compare it before and after tuning.

Dataset Overview

Dataset size

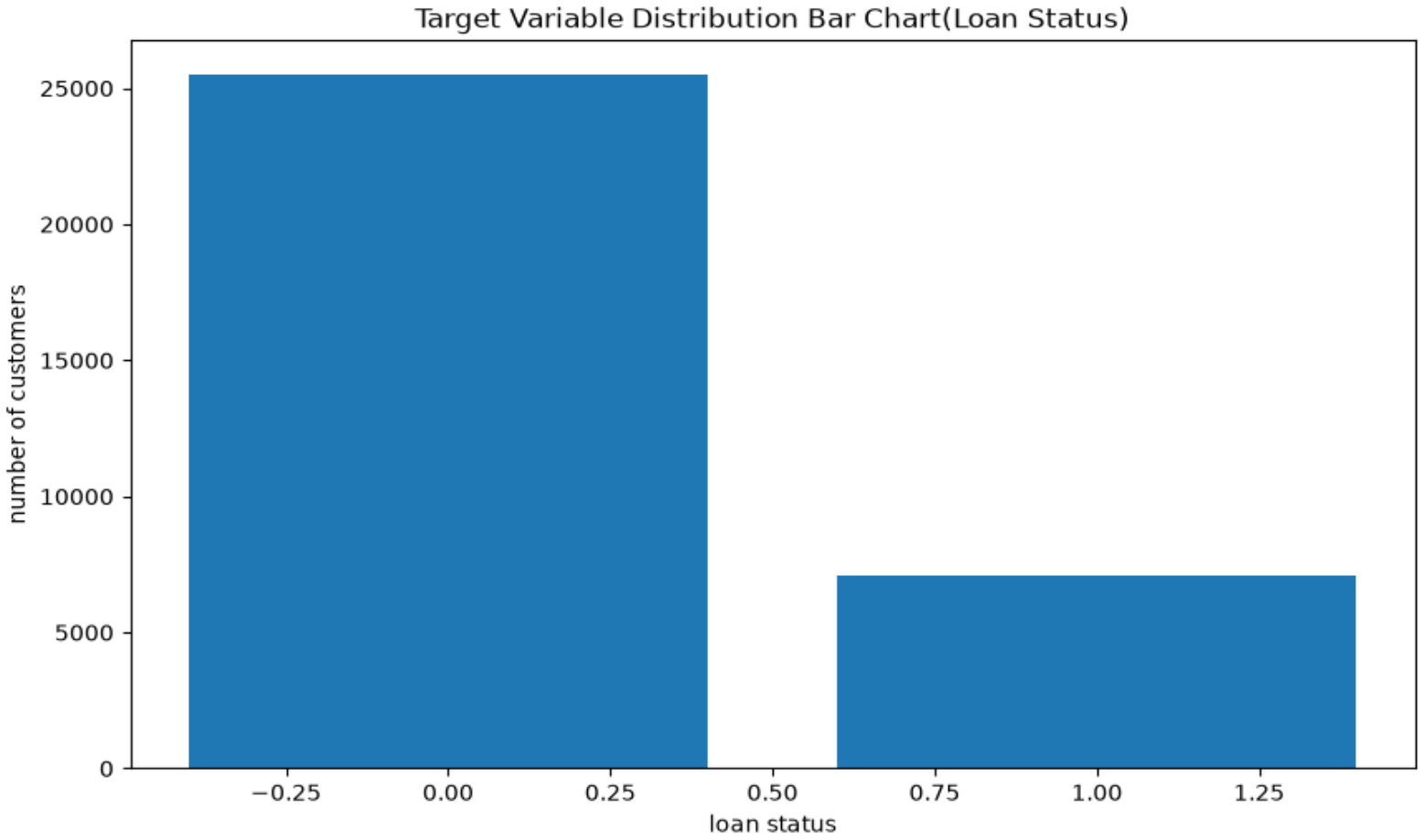
32,581 rows and 12 original columns.

Target variable

loan_status: 0 = negative class, 1 = positive class (default).

Class distribution

25,473 class-0 records and 7,108 class-1 records.

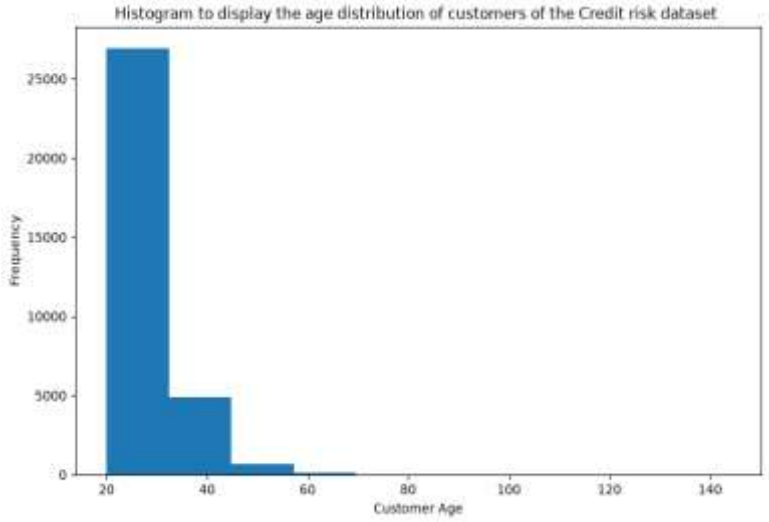


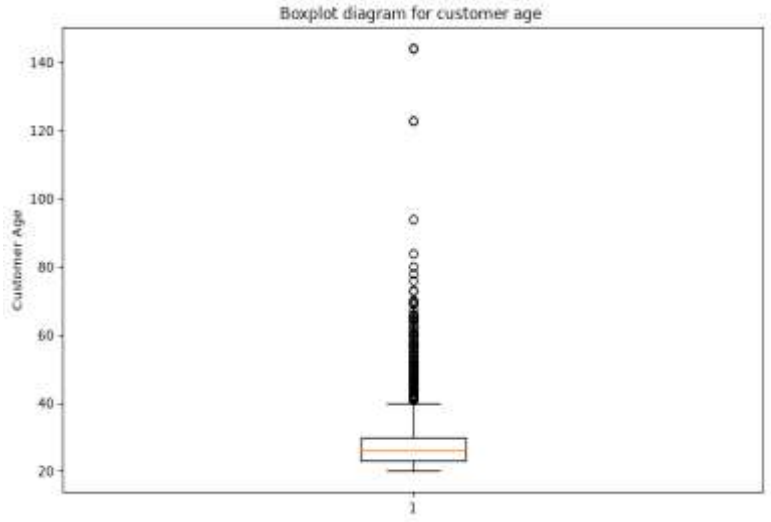
Important features: age, income, employment length, loan amount, interest rate, loan grade, loan intent and credit history length.

Data Issues Identified – EDA

Missing values: 895 in person_emp_length and 3,116 in loan_int_rate.

Outliers are visible in person_age; the maximum age in the raw data is 144.







Histogram

Most customer ages are concentrated in the younger adult range; the right tail contains fewer older customers.

Boxplot

Shows the median, middle 50% (IQR), whiskers and high-value outliers.

Correlation

The strongest visible relationship is age vs credit-history length (0.86); correlation shows association, not causation.

Preprocessing Summary

Step	What I did	Why
Missing values	Median imputation	Median is less affected by extreme values than the mean.
Binary encoding	Y/N → 1/0	Used because the variable has only two categories.
Ordinal encoding	loan_grade A–G → 1–7	Used because the grades have a natural order.
One-Hot Encoding	Home ownership + loan intent	Used because these nominal categories have no ranking.
Feature scaling	Min-Max 0–1	Prevents large-valued features from dominating distance-based methods and PCA.
Train/test split	80% / 20%	Training data learns the pattern; testing data checks performance on unseen data.

After encoding, the working dataset contained 20 columns (19 features + loan_status).

PCA – Principal Component Analysis

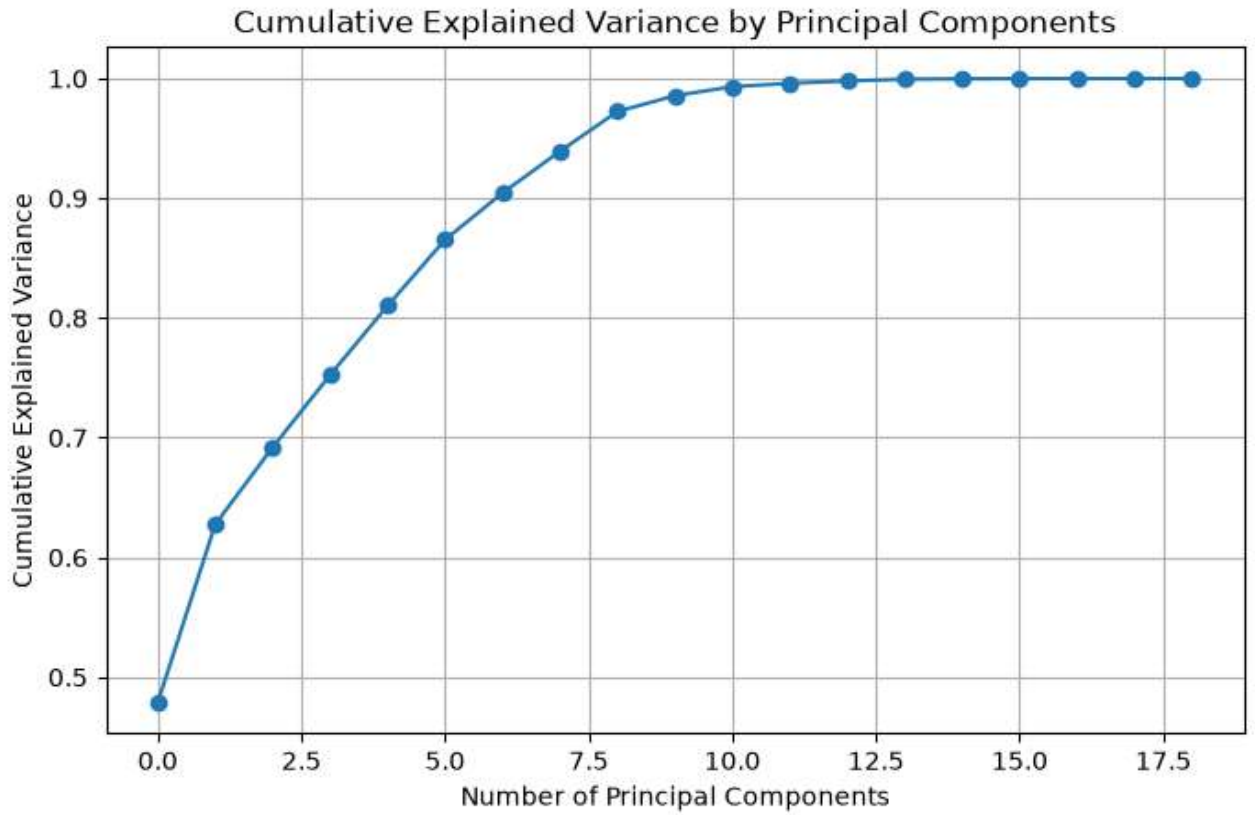
PCA reduces dimensionality by creating new variables called principal components.

Each component explains part of the total variance in the data.

The cumulative variance shows how much information is retained as more components are included.

The assignment required at least 90–95% variance; 8 components gave 93.94%.

Point	Variance
PC1	47.89%
PC2	62.75% cumulative
PC7	90.44% cumulative
PC8	93.94% cumulative

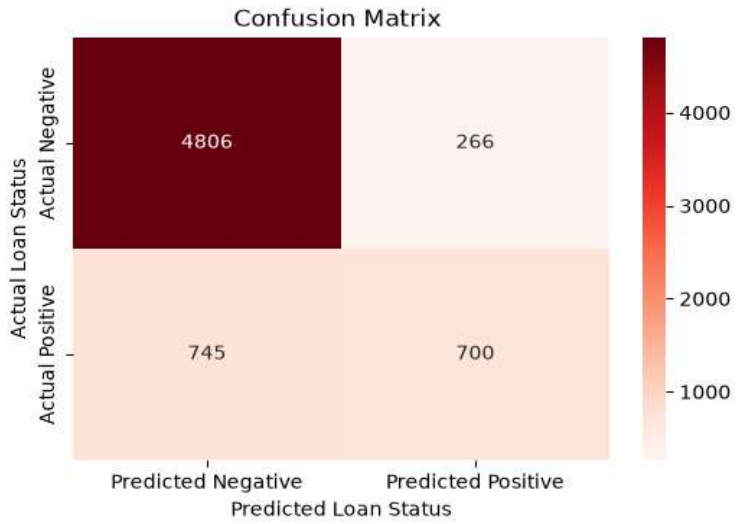


Result: 19 encoded features were reduced to 8 PCA features while retaining about 93.94% of the variance.

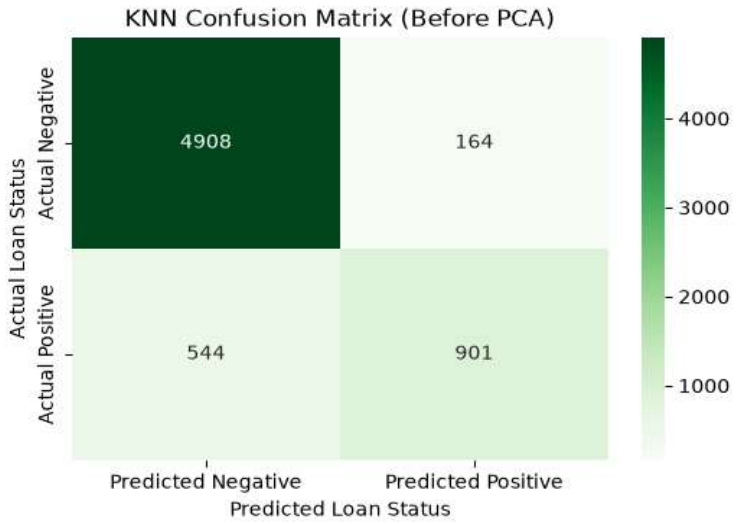
Model Comparison – Before PCA

Model	Accuracy	Precision	Recall	F1
Logistic Regression	84.49%	0.72	0.48	0.58
KNN	89.14%	0.85	0.62	0.72
Decision Tree	89.18%	0.74	0.78	0.76

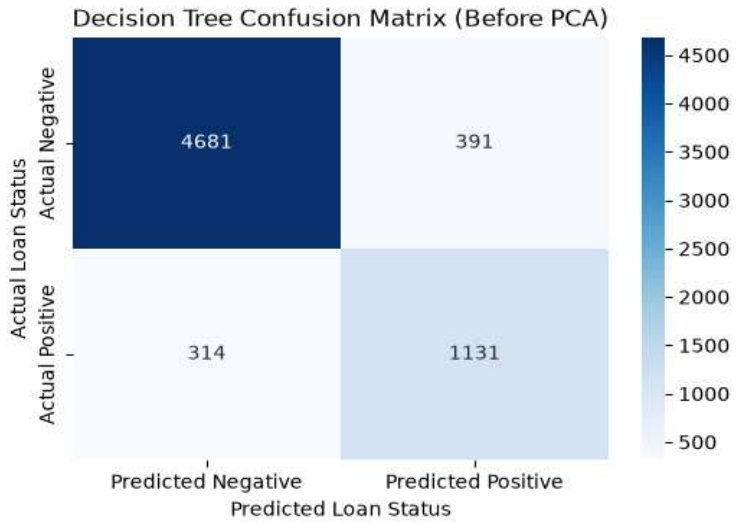
* Precision, recall and F1-score are for class 1 (default), because detecting default cases is the key business focus.



Logistic Regression: 700 true positives and 745 false negatives.



KNN: 901 true positives and 544 false negatives.



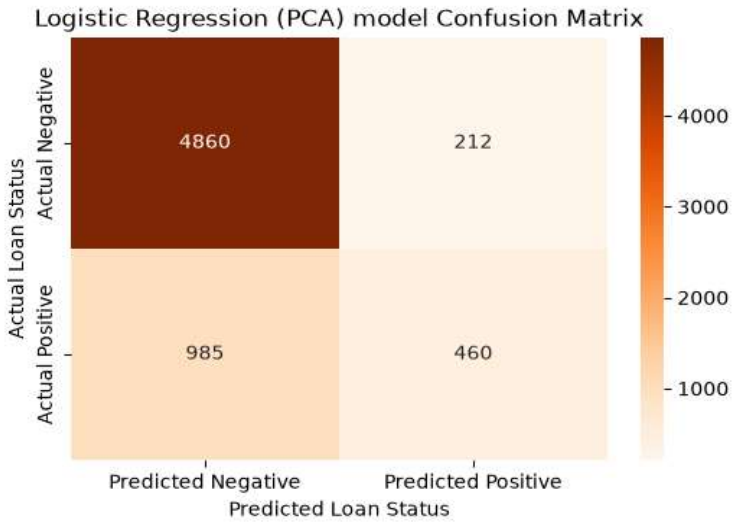
Decision Tree: 1,131 true positives and 314 false negatives.

Original-feature result: Decision Tree had the highest accuracy (89.18%) and highest class-1 recall (0.78) of the three.

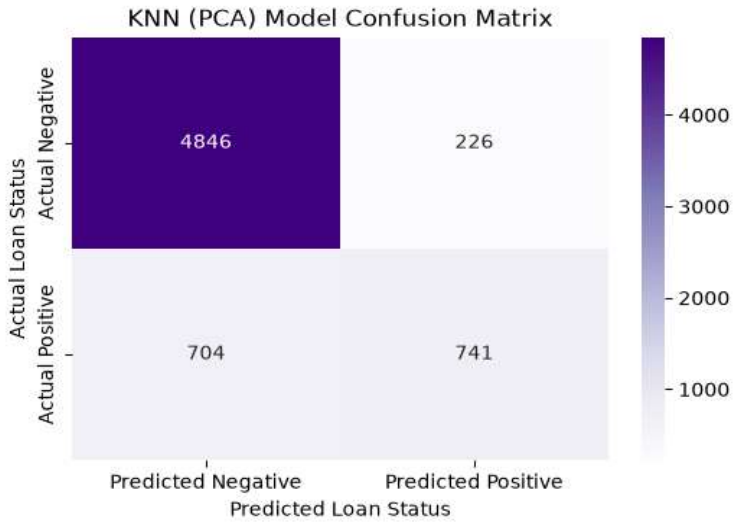
Model Comparison – After PCA

Model	Accuracy	Precision	Recall	F1
Logistic Regression	81.63%	0.68	0.32	0.43
KNN	85.73%	0.77	0.51	0.61
Decision Tree	84.67%	0.65	0.66	0.66

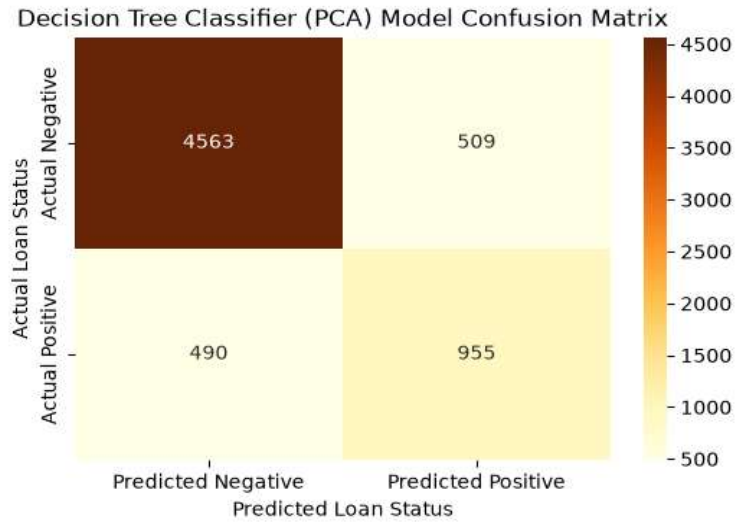
* Class-1 metrics (default). PCA used the 8 retained principal components.



Logistic Regression PCA: 460 true positives and 985 false negatives.



KNN PCA: 741 true positives and 704 false negatives.



Decision Tree PCA: 955 true positives and 490 false negatives.

PCA lowered accuracy for all three models in this run: LR 84.49→81.63%, KNN 89.14→85.73%, DT 89.18→84.67%.

Model Optimisation – GridSearchCV

Why this model?

The original-feature Decision Tree had the strongest pre-tuning accuracy (89.18%) and class-1 recall (0.78).

GridSearchCV settings

5-fold cross-validation; tested max_depth, min_samples_split, min_samples_leaf and criterion (Gini/Entropy).

Best parameter	Selected value
criterion	entropy
max_depth	10
min_samples_leaf	1
min_samples_split	2



Before vs after tuning

Accuracy: 89.18% → 93.36%. Class-1 precision: 0.74 → 0.97. Class-1 recall: 0.78 → 0.72.

Bias = error from a model that is too simple (underfitting). Variance = error from a model that is too complex (overfitting). GridSearchCV searches for a better balance.

Conclusion and Insights

The dataset was cleaned, encoded, scaled and split into training and testing data.

PCA reduced 19 features to 8 components while retaining 93.94% of the variance.

All three original-feature models outperformed their PCA versions in this experiment.

The original Decision Tree reached 89.18% accuracy and was selected for tuning.

GridSearchCV increased Decision Tree test accuracy to 93.36%.

The tuned model had very high class-1 precision (0.97), but recall remained 0.72, so false negatives still matter.

Best pre-tuning model

Decision Tree – Original Features
Accuracy: 89.18%
Class-1 recall: 0.78

Best tuned result

Decision Tree – GridSearchCV
Accuracy: 93.36%
Class-1 precision: 0.97

Business insight

False negatives are customers who actually defaulted but were predicted not to. This remains an important risk to monitor.

Overall takeaway: the original features contained useful information that was lost when reduced to 8 PCA components; tuning the Decision Tree then improved overall predictive accuracy.